

Advanced Electronic Circuits Spring 26

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HW1

P1) For a non-inverting amplifier with $R1 = 10 \text{ k}\Omega$ and $R2 = 400 \text{ k}\Omega$ calculate the value for a where the amplifier closed-loop gain will have an error of 10% as compared to if $a = \infty$.

$$A_{\text{real}} = V_o/V_i = 1/((1/a)+b) = 1/b * 1/(1+1/(ab))$$

$$\text{if } a=\infty, A_{\text{ideal}}=V_o/V_i = 1/b$$

$$b = R1/(R1+R2) = 10\text{k}/(10\text{k}+400\text{k}) = 1/41=0.02439$$

$$\text{where } 0.9 = 1/(1+1/(ab))$$

$$1=0.9(1+1/(ab))=0.9+0.9/(ab) ; .1=.9/(ab) ; .1(ab)=.9 ; ab=9 ; a=9/b=9/0.024=369$$

$$a=369$$

P2) Repeat the previous for a inverting amplifier.

$$A_{\text{real}}=V_o/V_i = 1/b * (1/ (1+ ((1-(1/b))/a))$$

$$\text{if } a=\infty, A_{\text{ideal}}=V_o/V_i = 1/b$$

$$b = -R2/R1 = -10\text{k}/400\text{k} = -0.025$$

$$\text{where } 0.9 = (1/ (1+ ((1-(1/b))/a))$$

$$a=369, \text{ same as before}$$

P3) Book Problem 1.27. Assume $a = \infty$

1.27 The audio panpot circuit of Fig. P1.27 is used to continuously vary the position of signal v_I between the left and the right stereo channels. (a) Discuss circuit operation. (b) Specify R_1 and R_2 so that $v_L/v_I = -1$ V/V when the wiper is fully down, $v_R/v_I = -1$ V/V when the wiper is fully up, and $v_L/v_I = v_R/v_I = -1/\sqrt{2}$ when the wiper is halfway.

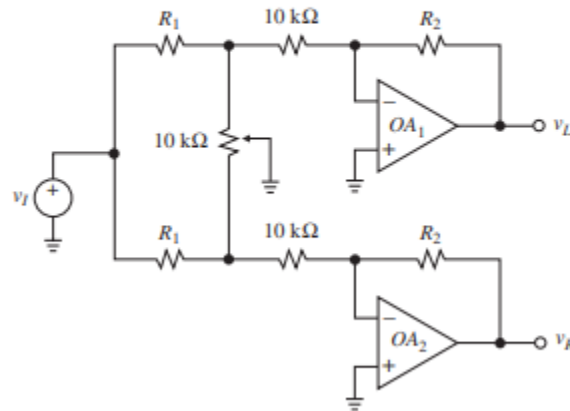


FIGURE P1.27

a) Discuss circuit operation

the left and right channels can be analyzed independently and symmetrically since the potentiometer and the voltage source create a division between them, so we'll just look at a single inverting amplifier. We'll do a system of equations with R_1 and R_2 in the two instances of $R_{pot} = 10k$ or $5k$, with $A = -1$ or $-1/\sqrt{2}$

b) find R_1 and R_2

V_n : both R_{pot} and R_3 are going to gnd, so voltage divider between those and V_{in}

$$V_n = V_{in} * (10k || R_{pot}) / (R_1 + (10k || R_{pot}))$$

$$i_{R3} = V_n / 10k = i_{R2} = -V_L / R_2$$

$$V_{in} * (10k || R_{pot}) / (R_1 + (10k || R_{pot})) / 10k = V_L / R_2$$

$$R_2 * (10k || R_{pot}) / (R_1 + (10k || R_{pot})) / 10k = V_L / V_{in}$$

$$R_{pot} = 10k, V_L / V_{in} = -1: R_2 * (10k || 10k) / (R_1 + (10k || 10k)) / 10k = -1$$

$$R_2 * 5k / (R_1 + 5k) = -10k; R_2 * 5k = -10k(R_1 + 5k); R_2 = -2(R_1 + 5k); R_2 = -2R_1 - 10k$$

$$R_{pot} = 5k, V_L / V_{in} = -1/\sqrt{2}: R_2 * (10k || 5k) / (R_1 + (10k || 5k)) / 10k = -1/\sqrt{2}$$

$$R_2 * 3.3k / (R_1 + 3.3k) = -10k/\sqrt{2}; R_2 * 3.3k = -10k/\sqrt{2} * (R_1 + 3.3k);$$

$$R_2 = -3/\sqrt{2} * (R_1 + 3.3k); R_2 = -3R_1/\sqrt{2} - 10k/\sqrt{2}$$

$$-2R_1 - 10k = -3R_1/\sqrt{2} - 10k/\sqrt{2}$$

$$R_1 = 24.142k\Omega; R_2 = 58.284k\Omega$$

P4) Book Problem 1.37. Assume $a = \infty$. Try to solve with only two op-amps; the solution is trivial when using 3 op-amps. The 3 op-amp solution will be accepted, but the problem will only receive half credit.

1.37 Design a two-input, two-output circuit that yields the sum and the difference of its inputs: $v_S = v_{I1} + v_{I2}$, and $v_D = v_{I1} - v_{I2}$. Try minimizing the component count.

note: non inverting sum,

subtractor gets $V_D = v_1 - v_2$

$V_S = v_1 + v_2 = 2v_1 - (v_1 - v_2) = 2v_1 - V_D$

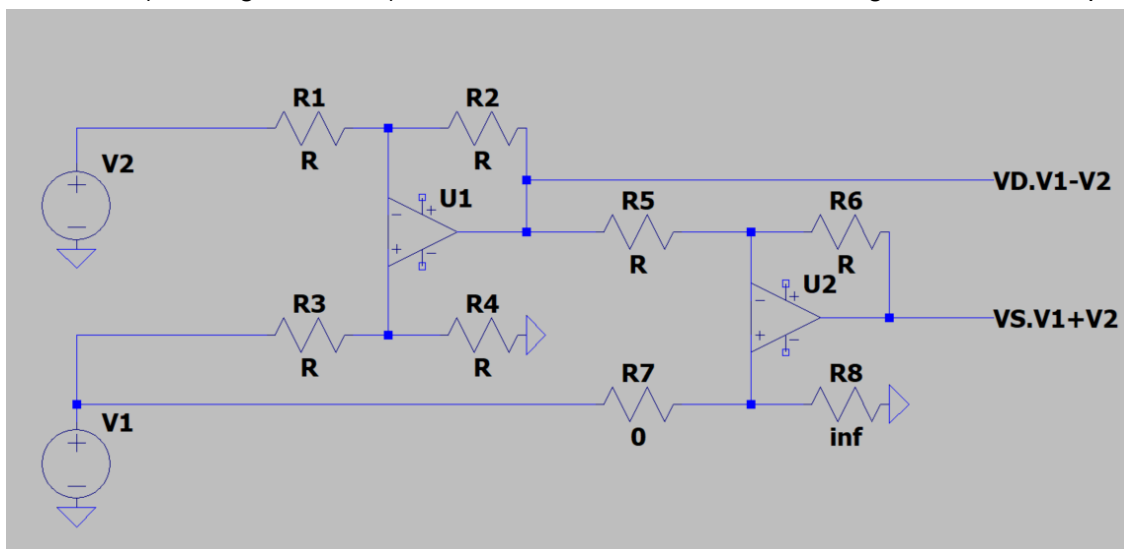
subtractor1 (v_1 on V_2+ , v_2 on V_1-): $V_{out} = -V_1(R_3/R_1) + V_2(R_4/(R_2+R_4))((R_1+R_3)/R_1)$

$V_D = -V_1 + V_2$, $R_3 = R_1$, $((R_1+R_3)/R_1) = 2$, $(R_4/(R_2+R_4)) = 0.5$, $R_1 = R_2 = R_3 = R_4$

subtractor2 (v_1 on V_2+ , V_D on V_1-): $V_{out} = -V_1(R_3/R_1) + V_2(R_4/(R_2+R_4))((R_1+R_3)/R_1)$

$V_S = -V_1 + 2V_2$, $R_3 = R_1$, $((R_1+R_3)/R_1) = 2$, $(R_4/(R_2+R_4)) = 1$, $R_2 = 0$

with $R_2 = 0$ (shorting V_2+ to v_1) no need for R_4 , $R_4 = \text{inf}$, all remaining resistors are equal



P5) For a generic negative feedback system with an error amplifier gain of $a\varepsilon = 50^3$ V/V and a closed loop gain of $A = 40$, find:

(i) The feedback gain, b , of the system,

$$A = a\varepsilon / (1 + a\varepsilon b)$$

$$1 + a\varepsilon b = a\varepsilon / A ; a\varepsilon b = a\varepsilon / A - 1 ; b = 1/A - 1/a\varepsilon = 1/40 - 1/50^3 = 0.024992$$

$$b = 0.024992$$

(ii) Find A for if $a\varepsilon$ decreases by 5% and 50%. Note that a drop of 50% is not considered a "small change."

$$A = a\varepsilon / (1 + a\varepsilon b)$$

$$5\%: b = 0.024992, a\varepsilon = .95 \cdot 50^3, A = 39.999326$$

$$50\%: b = 0.024992, a\varepsilon = .5 \cdot 50^3, A = 39.9872$$

(iii) Repeat the above assuming the initial value for $a\varepsilon$ was 50^2 V/V.

$$b = 1/A - 1/a\varepsilon = 0.0246$$

$$A = a\varepsilon / (1 + a\varepsilon b)$$

$$5\%: b = 0.0246, a\varepsilon = .95 \cdot 50^2, A = 39.9663$$

$$50\%: b = 0.0246, a\varepsilon = .5 \cdot 50^2, A = 39.370$$

P6) Book 1.49

1.49 A certain op amp has been configured as an inverting amplifier with two identical $10\text{-k}\Omega$ resistors, and is driven by a 1-kHz sinusoidal wave having $\pm 5\text{-V}$ peak values. Unfortunately, because of a fabrication error, the device exhibits $a = 10\text{ V/mV}$ for $v_O > 0$, but only $a = 2.5\text{ V/mV}$ for $v_O < 0$. Sketch and label v_I , v_O , and v_N versus time, and explain whether the device can still be used for the intended application.

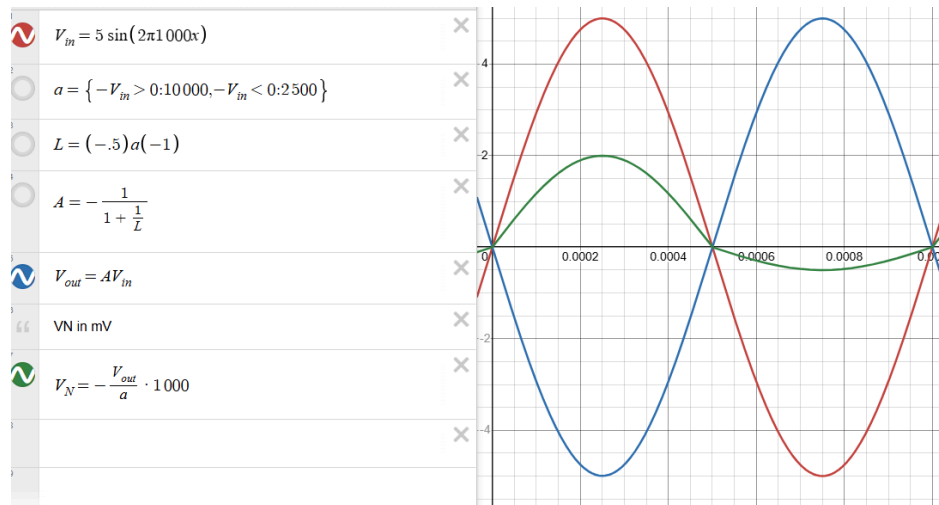
inverting op amp: $a_e = -a R_2/(R_1+R_2) = -0.5a$, $V_{out} = -aV_n$, $V_n = -V_{out}/a$

$V_{in} > 0$, $V_{out} < 0$, $a = 2500$, $a_e = -1250$, $b = -1$, $L = 1250$

$A = (-1) \frac{1}{(1+1/L)} = -0.9992$, $V_n = 2\text{mV}$

$V_{in} < 0$, $V_{out} > 0$, $a = 10000$, $a_e = -5000$, $b = -1$, $L = 5000$

$A = (-1) \frac{1}{(1+1/L)} = -0.9998$, $V_n = -0.5\text{mV}$



P7) Book 1.52

1.52 A certain audio power amplifier with a signal gain of 10 V/V is found to produce a 2-V peak-to-peak 120-Hz hum. We wish to reduce the output hum to less than 1 mV without changing the signal gain. To this end, we precede the power stage with a preamplifier stage with gain a_1 and then apply negative feedback around the composite amplifier. What are the required values of a_1 and b ?

$$x_o = A(x_i + x_n)$$

add pre amp, do some of the amplification earlier before noise gets in

$$x_o = A(a(x_i - b x_o) + x_n)$$

$$x_o = A a (x_i - b x_o + x_n/a)$$

$$x_o = 2\text{V} = A x_n, A = 10, x_n = 2/10 = .2\text{V}$$

$$x_o = 1\text{mV}, x_n = .2\text{V}, x_o/x_n = .001/.2 = .005 = A/(1+A a b)$$

$$x_o/x_i = 10 = A a (1+A a b)$$

$$a = 10/0.005 = 2000\text{V/V}$$

$$10 = 10 \cdot 2000 / (1 + 10 \cdot 2000 \cdot b), b = 0.09995$$

$$a = 2000\text{V/V}$$

$$b = 0.09995$$